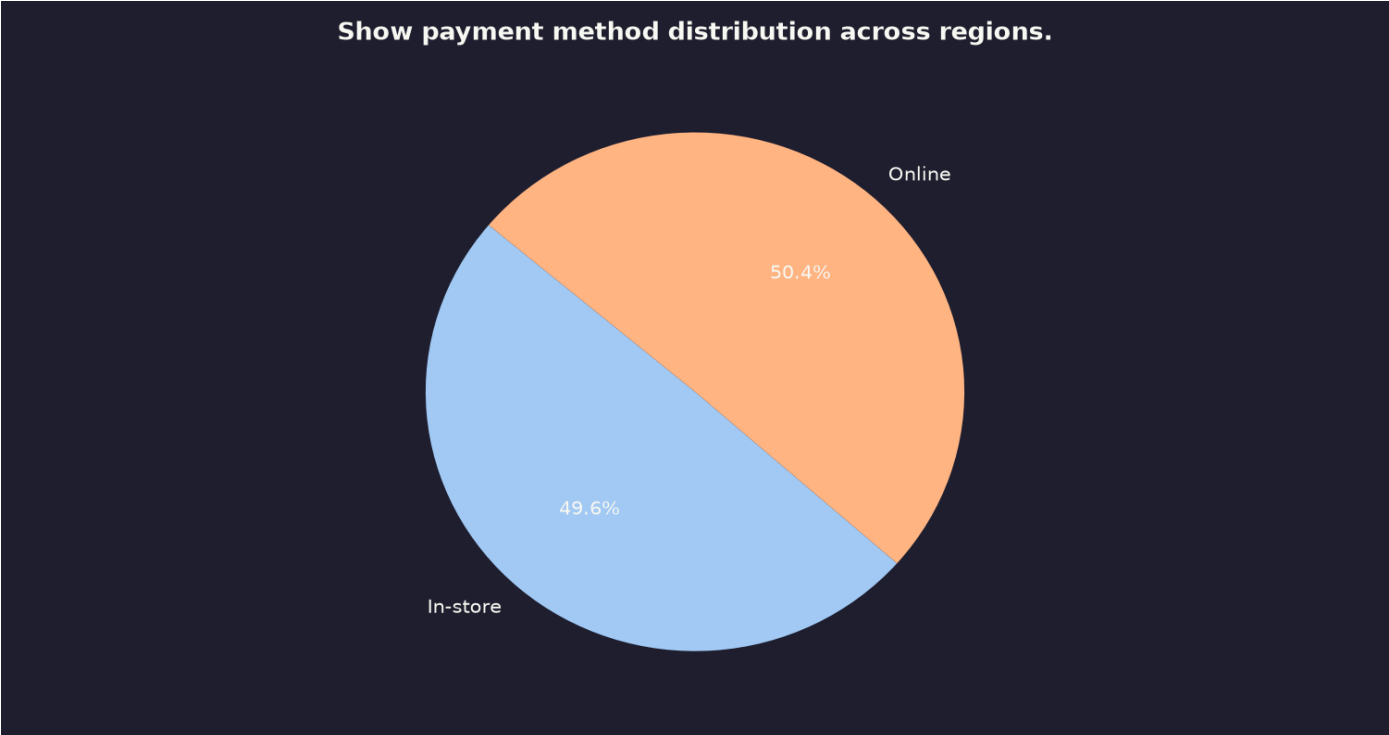


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** “Show payment method distribution across regions.”
- **Answer:** Both **In-store** and **Online** regions exhibit a balanced mix of payment methods. Cash is the most frequently used method in each region (~34 % of transactions), while Credit Card and Digital Wallet each capture roughly 32-33 % of the transactions. No region shows a dominant payment preference.

2. Dataset Overview & Data Quality Profile

Aspect Details
----- -----
Rows × Columns 12,575 × 11
Duplicates 0 (none found)

| **Missing Values** | **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, **Discount Applied** contain missing entries |

| **Potential Date Columns** | **Transaction Date** stored as string |

| **General Quality** | Dataset validated after duplicate removal; no structural issues beyond the missing values noted above. |

3. Data Cleaning & Validation Summary

- **Cleaning Step:** Removed duplicate rows (none existed) and created a clean copy.

```
result = df.drop_duplicates().copy()
```

- **Validation:** All 12,575 rows retained; dataset ready for analysis.

4. User Query & Analytical Approach

- **User Query:** “Show payment method distribution across regions.”

- **Analytical Methodology:**

1. **Group** the data by **Location** (region) and **Payment Method**.
2. **Count** the number of transactions per combination.
3. **Pivot** the counts into a matrix (**Location** rows, **Payment Method** columns).
4. **Calculate** row-wise totals and percentages for a clear distribution view.

5. Python Analysis Code & Executed Data Output

Code

```
# Count of transactions per payment method for each region (Location) distribution = (  
df.groupby(['Location', 'Payment Method']) .size() .unstack(fill_value=0) ) # Add total  
transactions per region distribution['Total'] = distribution.sum(axis=1) # Compute percentage  
share per payment method percentage = distribution.div(distribution['Total'],  
axis=0).applymap(lambda x: f"{x:.1%}") distribution.reset_index(inplace=True) distribution
```

Executed Output (as a table)

| Location | Cash | Credit Card | Digital Wallet | Total |

|-----|-----|-----|-----|-----|

| In-store | 2,138 | 2,020 | 2,063 | 6,221 |

| Online | 2,172 | 2,101 | 2,081 | 6,354 |

Percentage view (optional):

| Location | Cash | Credit Card | Digital Wallet |

|-----|-----|-----|-----|

| In-store | 34.4 % | 32.5 % | 33.1 % |

| Online | 34.2 % | 33.1 % | 32.8 % |

6. Key Insights & Strategic Recommendations

- **Balanced Payment Mix:** Both regions show a near-even split among cash, credit cards, and digital wallets. This suggests customers are comfortable using any of the three methods.
- **Cash Slight Edge:** Cash holds a modest lead (~34 %) in both channels. Retail operations should continue to support cash handling efficiently.
- **Digital Wallet Growth Opportunity:** While digital wallets already capture ~33 % of transactions, they are slightly behind cash and credit cards. Targeted promotions (e.g., discounts for wallet users) could nudge a higher share.
- **Channel-Specific Tactics:**
 - *In-store:* Emphasize quick cash checkout lanes and staff training for cash handling.
 - *Online:* Highlight secure credit-card processing and seamless digital-wallet integration to maintain the current balance.
- **Monitoring:** Set up a monthly dashboard tracking payment-method percentages to detect shifts (e.g., emerging preference for contactless payments).

Prepared by the Autonomous AI Data Analyst

![Show payment method distribution across regions. Chart](analysis_chart.png)